

Reinforcement Learning from Human Feedback

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1. Introduction

Reinforcement Learning from Human Feedback (RLHF) has become a key technique for aligning AI behavior with human preferences, especially in complex tasks where explicit reward functions are hard to define [1]. In this project, we evaluate RLHF methods in benchmark environments from OpenAI Gym, including CartPole and Acrobot.

Our objective was to generate preference-based trajectories by contrasting high- and low-performing policies, and to evaluate the performance of two RLHF methods: Direct Preference Optimization (DPO) and Proximal Policy Optimization with Reinforcement Learning from Human Feedback (PPO-RLHF).

To assess the impact of human feedback on policy learning, we systematically varied key hyperparameters—including dataset size, feedback frequency, and environment complexity—and analyzed how these variations influenced performance, stability, and the overall effectiveness of each RLHF approach.

2. Background and related work

• PPO-RLHF

2-step algorithm [2] :

1. Train reward function :

$$\hat{r}_\phi = \arg \max_{r \in R} \sum_{i=1}^K \log \left(\sigma \left(r(\tau_i^+; \phi) - r(\tau_i^-; \phi) \right) \right)$$

here, σ is the sigmoid function

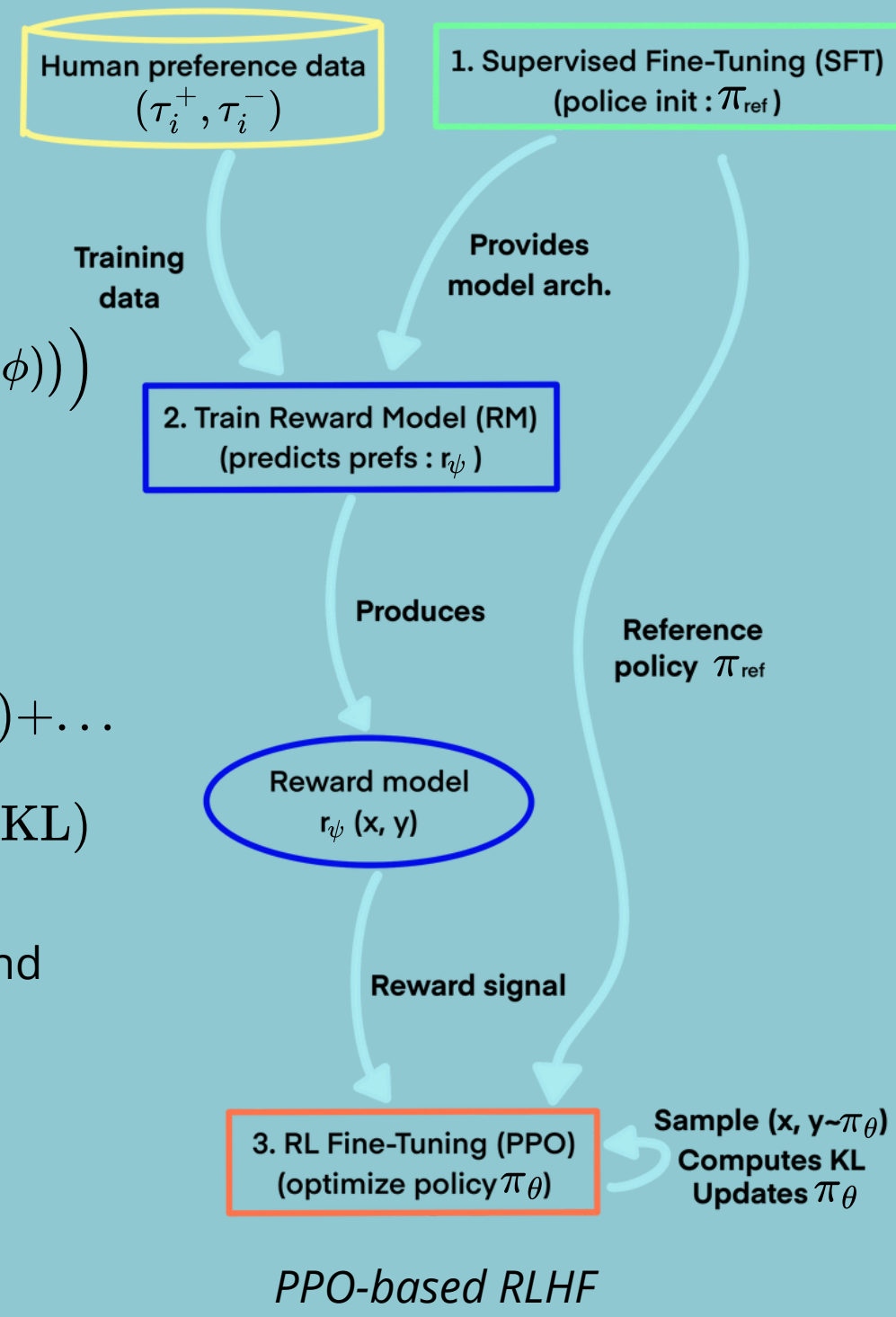
2. Optimize the policy using the reward model:

$$\arg \max_{\theta} L_t^{\text{CLIP} + \text{VF} + \text{S}}(\theta) = \hat{\mathbb{E}}_t \left[L_t^{\text{CLIP}}(\theta) + \dots \right. \\ \left. \dots - c_1 L_t^{\text{VF}}(\theta) + c_2 S[\pi_\theta](s_t) \right] - \mathbb{E}(\beta \cdot \text{KL})$$

L_t^{CLIP} is used to limit policy updates and ensure stable learning

PPO aims to balance :

- Easy implementation
- Sample efficiency
- Reliable performance.



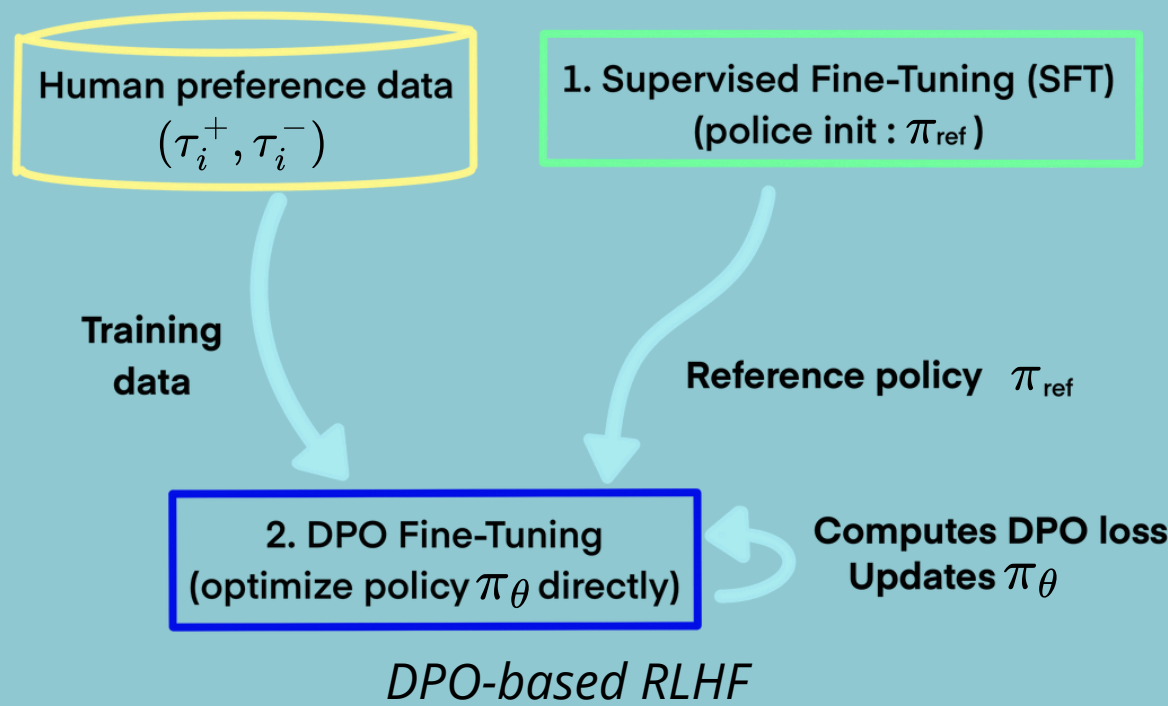
• Direct Preference Optimization (DPO) [3]

End-to-end algorithm (simplified pipeline) :

- DPO directly optimizes policies towards human feedback trajectories, eliminating explicit reward modeling. This is done by maximizing the [Binary Logistic Loss](#).

This loss encourages the model to prefer trajectories aligned with human preferences.

$$\ell_{\text{DPO}}(\theta) = -\frac{1}{K} \sum_{i=1}^K \log \left(\sigma \left(\beta \cdot \log \left(\frac{\pi_\theta(\tau_i^+)}{\pi_{\text{ref}}(\tau_i^+)} \right) - \beta \cdot \log \left(\frac{\pi_\theta(\tau_i^-)}{\pi_{\text{ref}}(\tau_i^-)} \right) \right) \right)$$



Advantages of DPO :

- Simpler implementation
- Reduce risk of reward overfitting
- Improves stability

Disadvantage of DPO :

- Lack of explicit reward model

• Motivation for comparison

Comparing PPO-RLHF and DPO on the same environment using identical preference datasets, allows us to evaluate the trade-offs between:

- **Pipeline complexity:** DPO has fewer moving parts, while PPO-RLHF introduces additional sources of error through the reward model.
- **Stability:** DPO is expected to exhibit more stable training behavior.
- **Performance:** PPO-RLHF may benefit from an interpretable reward signal, which could lead to better generalization in some settings.

5. Conclusion

Our comparison between PPO-RLHF and DPO reveals several key insights. As expected, DPO offers a simpler and more streamlined approach, with fewer hyperparameters and easier implementation. This simplicity contributes to more stable training behavior.

On the other hand, because of its large hyper-parameter count, PPO-RLHF is harder to tune, even with an interpretable reward model. More research is required to reach performance on par with DPO in these two environments.

Importantly, both algorithms showed improved policy quality as the dataset size n increased, confirming the value of large-scale human preference data. Ultimately, the choice between PPO-RLHF and DPO involves a trade-off between simplicity and interpretability, with performance depending on the specific task and available resources [4].

3. Approach

Two parts :

1. Trajectory Generation :

Pairs of trajectories are generated using two policies with different reward levels (train by REINFORCE), assigning preference probabilities based on their relative rewards. Preference is then sampled probabilistically to build the dataset for training.

$$P(\tau_1 \text{ is preferred}) = \frac{\exp(R(\tau_1))}{\exp(R(\tau_1)) + \exp(R(\tau_2))}$$

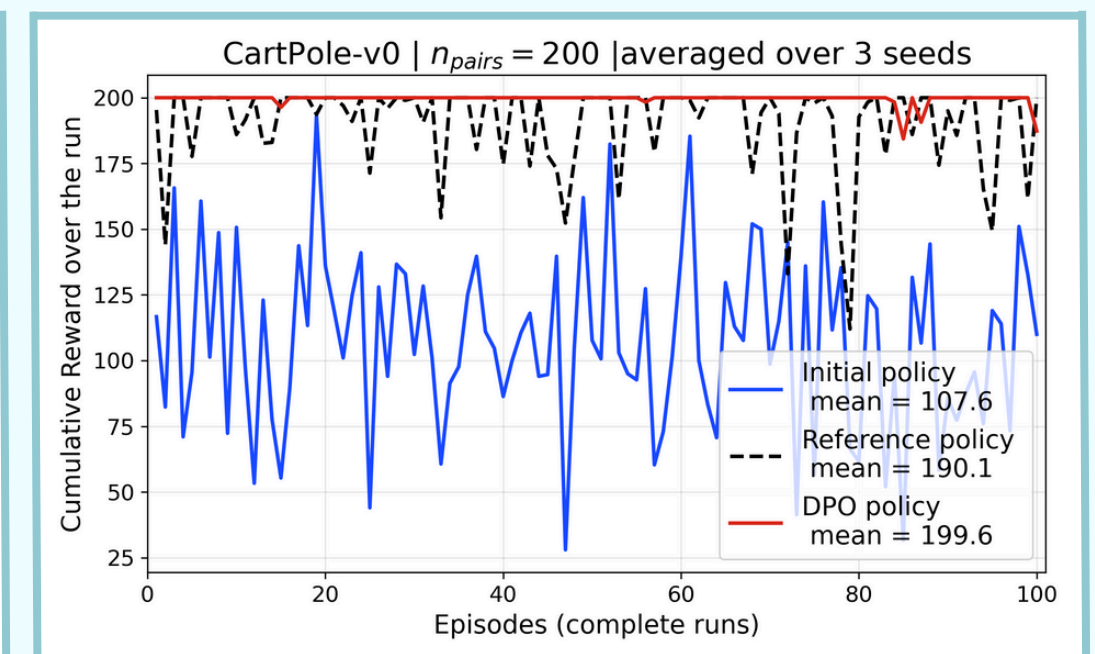
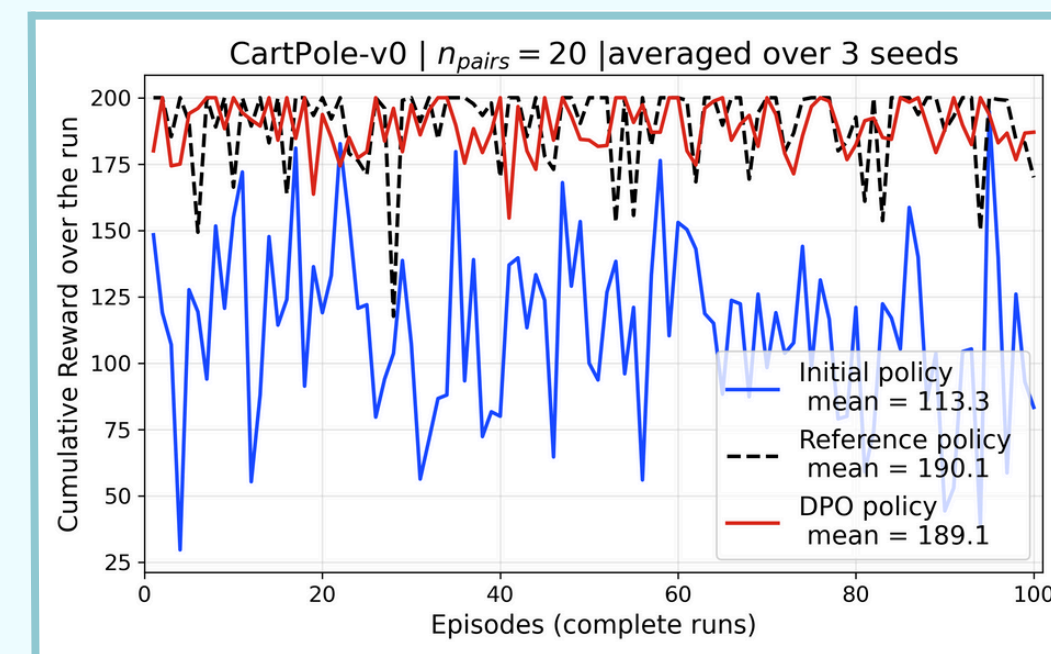
2. RLHF algorithm application :

Use DPO and PPO-RLHF to generate a third policy and compare their performances in terms of reward optimization, policy improvement, and alignment with preferences. Evaluate how each algorithm integrates feedback to enhance learning.

4. Results

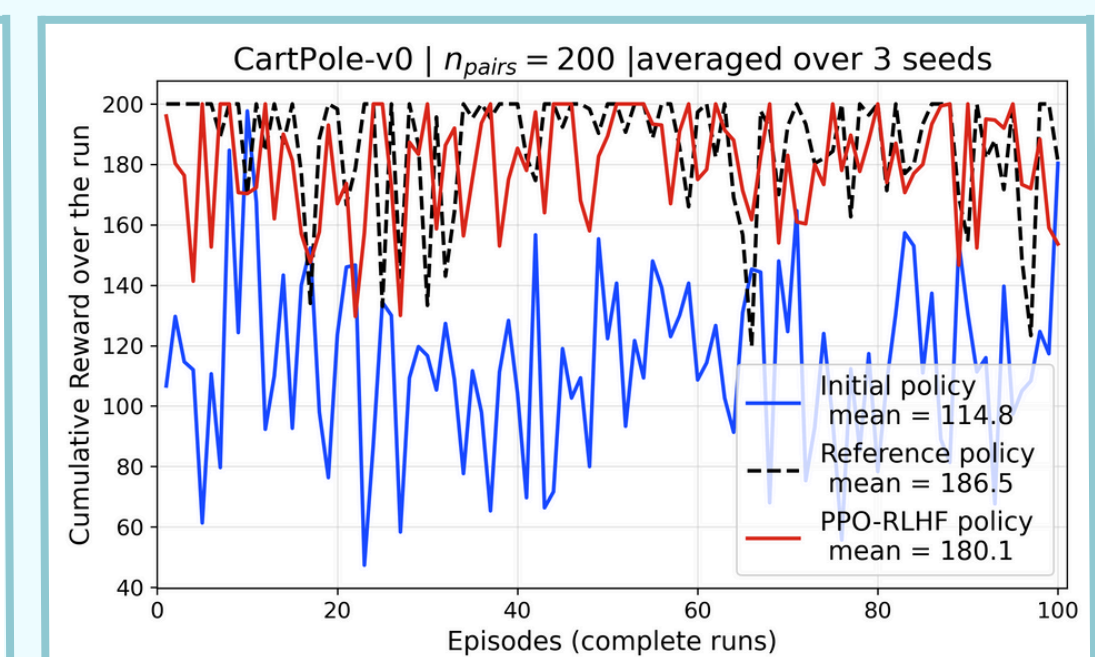
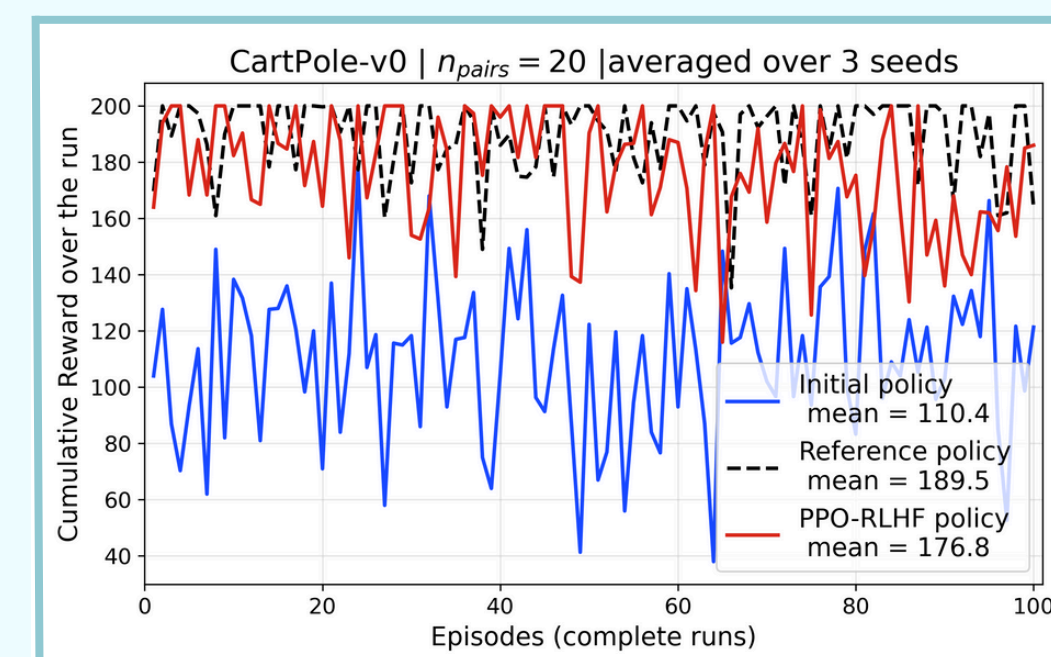
• CartPole

DPO results :



DPO cumulative reward compared to the initialisation and reference ones, for two sizes n of the preference dataset

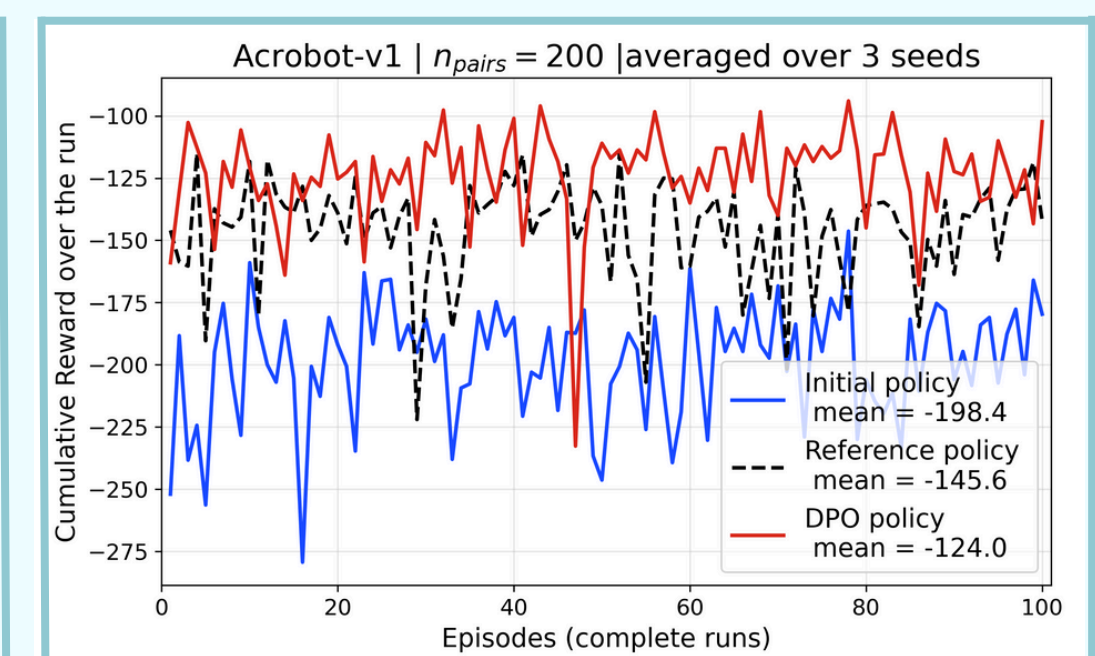
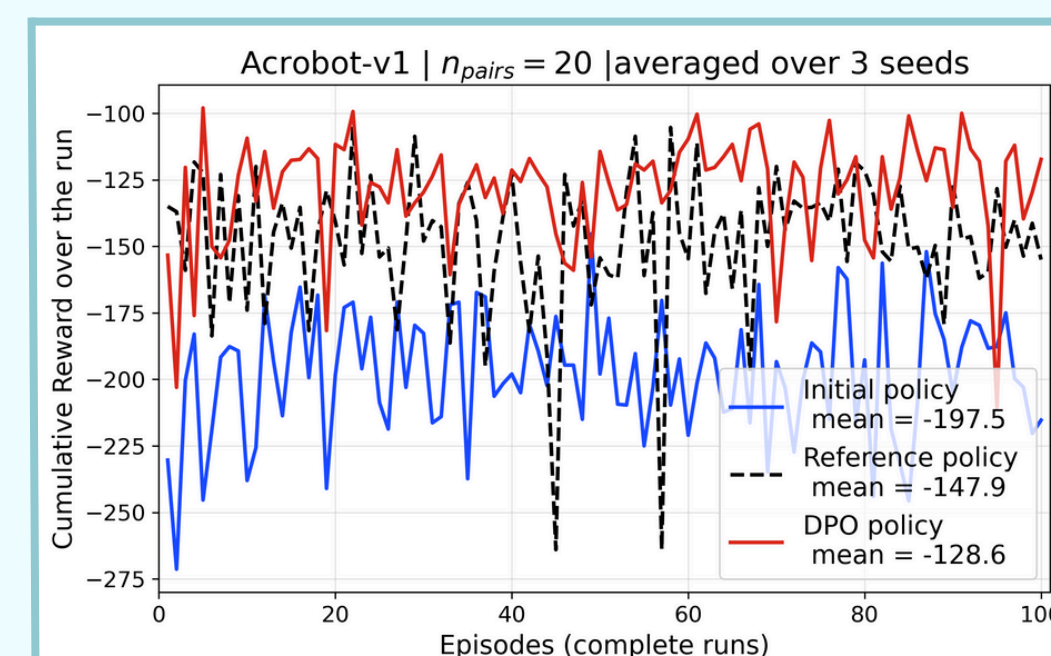
PPO-RLHF results :



PPO-RLHF cumulative reward compared to the initialisation and reference ones, for two sizes n of the preference dataset

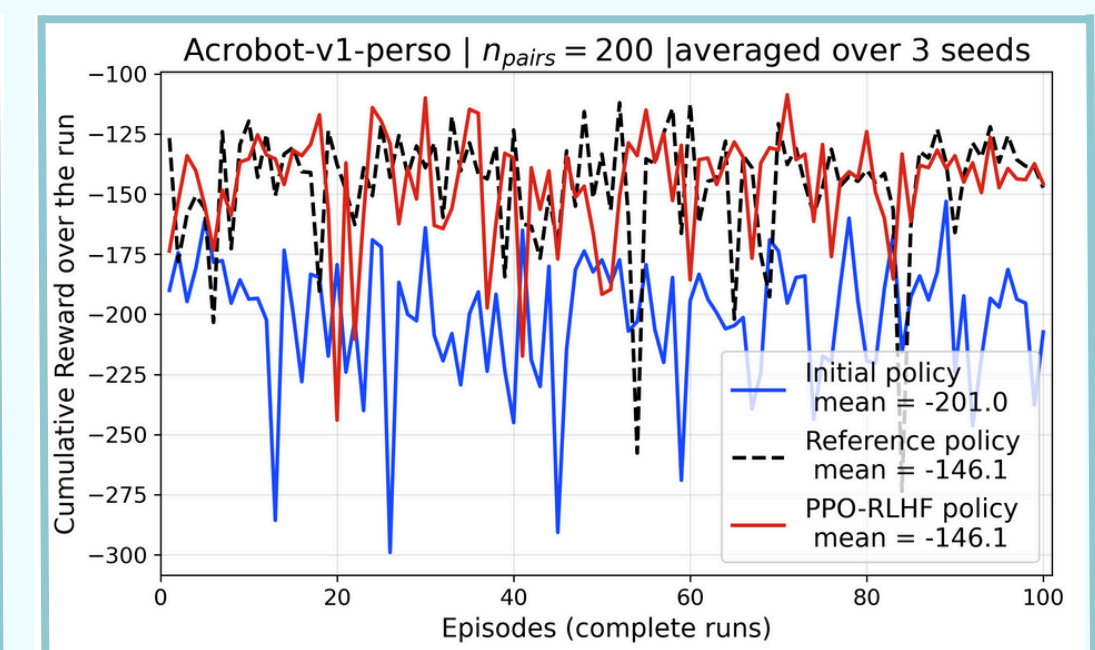
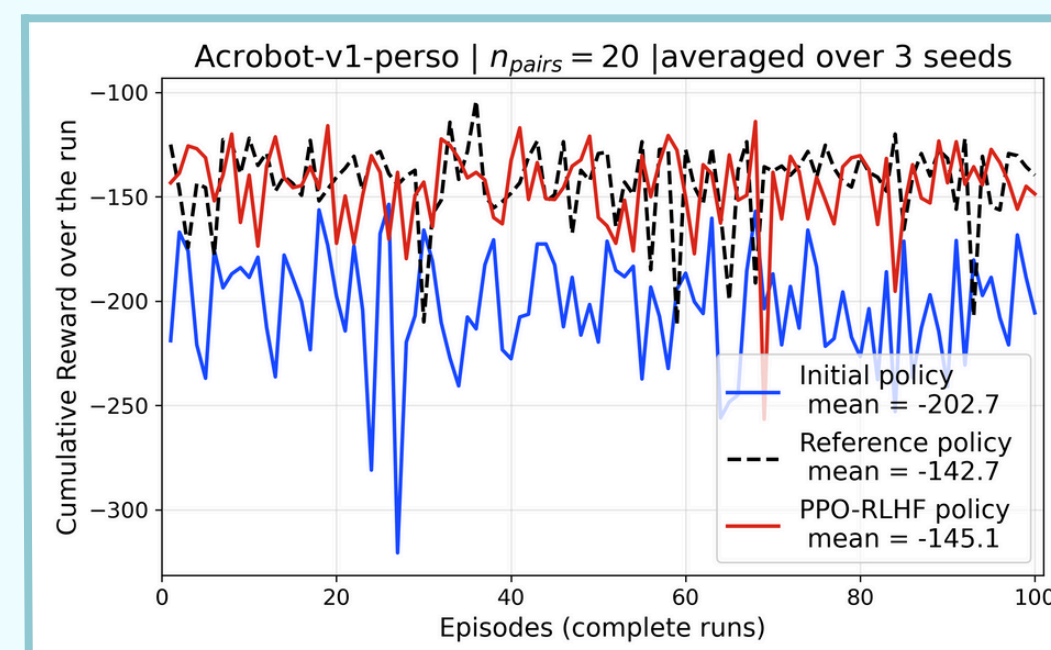
• Acrobot

DPO results :



DPO cumulative reward compared to the initialisation and reference ones, for two sizes n of the preference dataset

PPO-RLHF results :



PPO-RLHF cumulative reward compared to the initialisation and reference ones, for two sizes n of the preference dataset

REFERENCES

- [1] Reinforcement Learning from Human Feedback. Wikipedia
- [2] Zheng, Rui, et al. *Secrets of RLHF in Large Language Models Part I: PPO*. 2023
- [3] Rafailov, Rafael, et al. *Direct Preference Optimization: Your Language Model Is Secretly a Reward Model*. 2024
- [4] Xu, Shusheng, et al. *Is DPO Superior to PPO for LLM Alignment? A Comprehensive Study*. 2024